

INVESTIGATION REPORT

Issue 1009: Nested hardware-loop end addresses too close together

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1009
Disposition	Waiver, with a stated software rule
Record date	2026-09-09
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1009
Issue filed	2024-06-24
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL master 6033d2b, COREV_PULP=1 FPU=0
Disposition	Waiver, with a stated software rule
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	4 cited, 3 present, 1 not retained

Disposition meaning. The code is unreachable only while software obeys an explicitly stated rule. The rule is part of the disposition and is recorded in the Action section.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The uncovered branch at cv32e40p_controller.sv:831 (is_hwlp_body==1 while in DECODE_HWLOOP with hwlp_end_addr[1]==pc_id+8 and hwlp_counter[1]<=1) requires HWLoop0 to be active (counter0>1, start0<=pc, end0>=pc+4) at the moment HWLoop1's end register equals pc+8. That forces end0 >= end1-4: loop1's end sits inside loop0's body. HWLoop0 is the inner loop and HWLoop1 the outer, and the PULP HWLoop rule requires the outer end to be at least two 32-bit instructions past the inner end (end1 >= end0+8), which makes the branch unsatisfiable (z3 unsat with the invariant, sat without it). The hardware does not validate start/end/count relationships, so this cannot be proven from RTL: a bounded formal cover reaches the branch in 8 cycles with an illegally nested register set (end1=0x10 inside loop0 [0x4,0x2c], counter1=0). The mirror branch at line 840 (loop0 ends inside an active loop1 body) is the legal nesting and is covered, which explains the asymmetry. Route: waiver with the stated software invariant. Caveat lowering confidence: the formal witness has counter1=0, so a stale HWLoop1 end register left over from a finished loop and later enclosed by a fresh HWLoop0 body would also reach the branch with sequential (non-nested) loops; the waiver invariant must be read as a register-content invariant (whenever HWLoop0 is active, hwlp_end_addr[1] must not lie in [start0, end0+4]), and a human should consider a directed test on a PULP-enabled testbench before waiving. Behaviour in either arm is benign (stays DECODE_HWLOOP or returns to DECODE).

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_controller.sv:831 branch is_hwlp_body==1: reachable only if hwlp_end_addr[1] (HWLoop1, the outer loop) lies inside an active HWLoop0 body (end0 >= end1-4). The CV32E40P HWLoop programming rules require HWLoop0 to be nested inside HWLoop1 with the outer end at least two 32-bit instructions past the inner end (end1 >= end0+8); under that rule the branch is unsatisfiable (z3 unsat, runs/v1-cv32e40p-1009/smt/with_invariant). Hardware does not enforce the rule (bounded cover reaches the branch with an illegal setup, runs/v1-cv32e40p-1009/formal/c_line831). Waive the is_hwlp_body==1 arm at line 831 as software-illegal, with the invariant: whenever hwlp_counter[0] > 1, hwlp_end_addr[1] is not in [hwlp_start_addr[0], hwlp_end_addr[0]+4] (this also covers stale HWLoop1 registers). Line 840 is the legal mirror and remains covered.

Checker used

checker_1009.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	smt-model	smt/with_invariant.out	present	z3 unsat: line-831 condition + is_hwlp_body=1 impossible under nesting invariant end1>=end0+8, start0>=start1 (no wraparound, 4-byte aligned)
E2	smt-model	smt/without_invariant.out	present	z3 sat without the invariant; model is the illegal case: end1=pc+8, cnt1=0, pc inside loop0 body with cnt0>1
E3	proof	formal/c_line831/sby.log	present	smtbmc cover PASS, reached step 8: hardware alone (unconstrained program memory) can reach the exact line-831 branch with is_hwlp_body=1
E4	counterexample	formal/c_line831/job/engine_0/trace0.vcd	not kept	cover witness: DECODE_HWLOOP, pc=0x8, loop0=[0x4,0x2c] cnt 0x100, loop1 end=0x10 cnt 0 -> loop1 end inside loop0 body; rendered in cover_line831.png

Paths are relative to runs/v1-cv32e40p-1009. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- smt-model. A satisfying assignment from the SMT solver for the stated query.
- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.

Artefact(s) E4 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- The waiver holds only while software obeys the rule stated in Action. If software can violate that rule, the waiver does not hold and the code is reachable.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a

- separate run.
- Public demonstration runs on open cores are not client engagements and not production verification results.

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Provenance and measurement

Run directory	runs/v1-cv32e40p-1009
Title source	Title and configuration read back from reports/issues/cv32e40p_issue_1009.pdf, the previously published report for this issue.
Inputs shown	title, body(links stripped), labels
Human record	Avinash 2026-09: waiver with SW nesting rule end1 >= end0+8; z3 UNSAT under rule, SAT (end1=end0+4) without
Assessed confidence	0.65. This is the investigator's own estimate. It is not a measurement and not a reliability figure.
Iterations	2

Generated by reporting/scripts/issue_document.py from runs/v1-cv32e40p-1009/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.